Experiment 16: Named Entity Recognition using SpaCy in Python

1. Aim

To write a Python program using the SpaCy library to perform Named Entity Recognition (NER) on a given text.

2. Procedure

1. Import the `spacy` library in Python.
2. Load the English language model using `spacy.load('en_core_web_sm')`.
3. Define a sample text containing named entities.
4. Process the text with `nlp(text)` and iterate over `doc.ents`.
5. Display each recognized named entity and its entity label.

3. Program

```
import spacy

nlp = spacy.load("en_core_web_sm")
text = "Google was founded in California in 1998."

doc = nlp(text)
entities = [(ent.text, ent.label_) for ent in doc.ents]

print("Named Entities:", entities)
```

4. Output

```
Named Entities: [('Google', 'ORG'), ('California', 'GPE'), ('1998',
'DATE')]
```

5. Result

Thus, the Python program was successfully executed to perform Named Entity Recognition (NER) on a given text using `spacy`.